

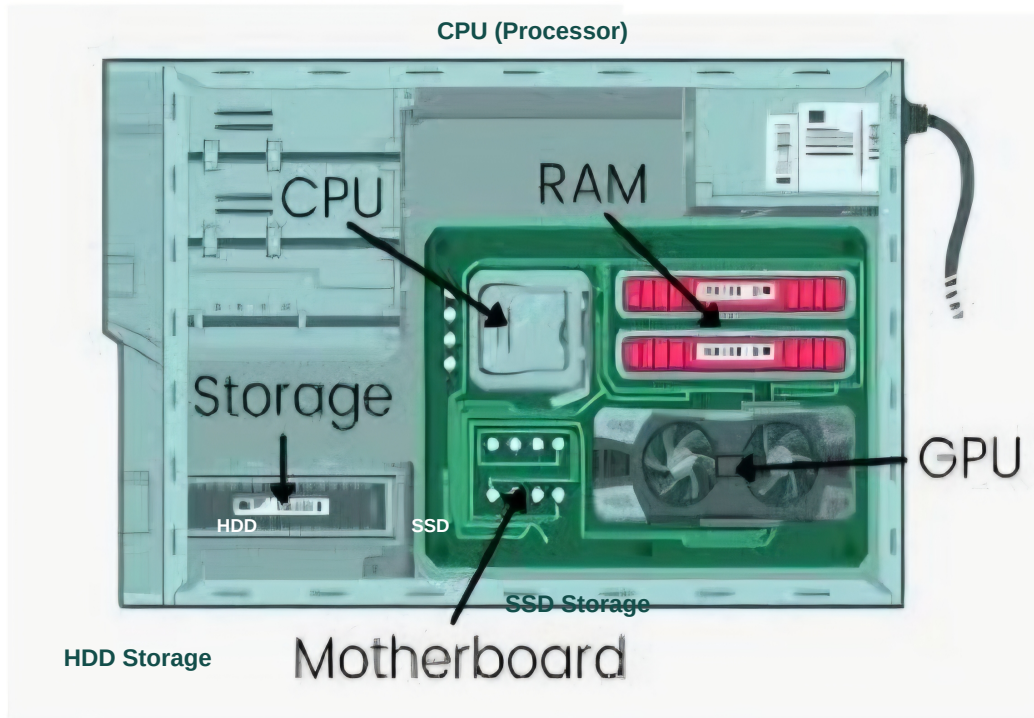
Overview of Hardware Components Theory

Motherboard • CPU • RAM • GPU • HDD & SSD Storage

Task 1 — Labeled Diagram of a Desktop PC

Below is a simplified diagram of what's inside a desktop PC case, with the motherboard, CPU, RAM, GPU, and both storage types labeled.

PC Case



Key: The motherboard (green) is the base that connects everything. The CPU sits in a socket near the top. RAM sticks plug in beside the CPU. The GPU is a card inserted lower down. The HDD and SSD are storage drives mounted in the case and connected to the motherboard by cables.

Task 2 — HDD vs SSD Comparison Table

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower — mechanical spinning disk and moving read/write head	Much faster — no moving parts, data read electronically
Durability	Less durable — moving parts can be damaged by shocks or drops	More durable — shock-resistant, no mechanical wear
Noise & Heat	Produces some noise and heat from the spinning disk	Silent operation, runs cooler
Price per GB	Cheaper — more storage for the same cost	More expensive — costs more per GB of storage
Typical Use Cases	Bulk storage: backups, media libraries, older desktops	Operating system drive, gaming PCs, laptops, fast boot times

Quick summary: SSD wins on speed and durability (great for a gaming PC or laptop that gets moved around). HDD wins on price-per-GB (good for a desktop that just needs cheap bulk storage, like a media/backup drive).

Task 3 — Role of CPU, RAM & GPU in Gaming (PUBG / Free Fire)

CPU (Central Processing Unit)

The CPU is the 'brain' of the PC — it runs the game's logic, physics, and AI calculations, such as tracking player positions and processing hit detection. In a game like PUBG, a faster CPU keeps the match running smoothly even when many players and objects are active on screen at once.

RAM (Random Access Memory)

RAM temporarily stores game data that needs to be accessed quickly, like the current map, textures, and character data, so the CPU and GPU don't have to keep fetching it from slower storage. More RAM lets a game like Free Fire load maps faster and switch between menus and gameplay without stuttering.

GPU (Graphics Processing Unit)

The GPU renders everything you see on screen — the 3D environment, character models, lighting, and effects like gunfire or explosions. A strong GPU lets PUBG or Free Fire run at higher graphics settings and a smoother, higher frame rate, which matters a lot in fast-paced shooting games.

Task 4 — Upgrading a PC to Run the Latest FIFA Smoothly

Modern FIFA titles are demanding on graphics and rely on a balanced system. Here's what to prioritize upgrading, and why:

- **GPU (Graphics Card):** The single biggest upgrade for FIFA. It directly determines frame rate, resolution, and how smooth crowds, player animations, and pitch details look.
- **RAM:** Upgrading to at least 16GB prevents stuttering when the game, background apps, and streaming/recording software all run at once.
- **Storage (switch HDD → SSD):** An SSD dramatically cuts loading times between matches, menus, and replays, and reduces in-game texture pop-in.
- **CPU:** If the current CPU is several generations old, it can bottleneck a new GPU. Worth upgrading if the rest of the system is modern but frame rates still feel capped.
- **Cooling / Power Supply (PSU):** A better GPU/CPU draws more power and heat, so the PSU wattage and case cooling may need checking to keep the upgraded parts running reliably.